

Report

SBA 928

Enhancing Market Research with AI Prompt Engineering

Local Model Fine-Tuning Resubmission

Richard Watson Jr.

October 6, 2026

Retail evidence, five task families, model comparison and safeguards

Prepared from the saved resubmission results for learner review

Contents

Section	Report page
Overview - objective and response to instructor feedback	2
Steps - dataset inspection and cleaning	3
Steps - customer purchasing and country sales	4
Steps - product demand and inventory decisions	5
Steps - monthly trends and competitor evidence	6
Steps - prompt engineering and dataset diversity	7
Steps - model loading and fine-tuning	8
Steps - comparison, scoring and error review	9
Steps - quantitative audit and safeguards	10
Conclusion	11
Appendix A - actual prompt variations	12
Appendix B - evidence and reproducibility	13

Repository: github.com/rwnycvet/sba928-market-research

Resubmission branch: [sba928-resubmission](#)

Evidence checkpoint: **3170416**, "Expand multitask fine-tuning and add audits and verified briefs." This report was prepared after that checkpoint.

Page numbers exclude the cover. The original submission is preserved under submission_v1/; earlier revision experiments remain under results_v2/.

Overview

Objective and response to instructor feedback

The research objective is to understand recorded customer purchasing patterns, product demand and differences in this retailer's sales across countries, so marketing and inventory decisions can be based on measured evidence.

The project combines Python analysis of the supplied Online Retail data with local FLAN-T5 fine-tuning. The resubmission expands the training tasks and documents quantitative evaluation while retaining the model's remaining failures.

Instructor feedback	Resubmission response
Country-only fine-tuning limited prompt diversity.	Expanded training to five balanced task families: consumer behavior, country coverage, market trends, product demand and competitor research.
Empty baseline responses and a loading warning required investigation.	Pinned Transformers 4.57.6 and a model revision; checked loading keys, configuration and embedding storage before training. Baseline response failures are still reported.
A more quantitative bias/fairness audit was needed.	Measured warning decisions on 16 controlled geographic-label cases, including missed and false warnings; evaluated Python mitigation separately.
The final Per Scholas report should accompany the repository.	Prepared this report in the supplied template structure, with evidence paths and reproducibility details.

Principal results

The v3 model matched the strict reference on **132/160 synthetic test prompts (82.5%)**. Its geographic-label audit remained **8/16 correct**; the Python warning safeguard was **16/16** on that defined rule. These are distinct results with different scopes.

AI assistance supported explanations, scaffolding, troubleshooting and report preparation. Training and analysis evidence comes from the project artifacts and recorded learner runs. The learner must review and be able to explain the submission.

Steps

1. Inspect and clean the supplied retail dataset

The supplied file contains **541,909 rows and eight columns**. Each row is a product entry, so several rows can belong to one invoice. The recorded period runs from December 1, 2010 through December 9, 2011.

Inspection finding in the original file	Rows
Missing CustomerID	135,080
Missing Description	1,454
Exact duplicates beyond the first occurrence	5,268
InvoiceNo beginning with C (cancellation rows)	9,288
Quantity at or below zero	10,624
UnitPrice at or below zero	2,517

Duplicate handling: Keep the first occurrence of each exact eight-column row. This is a documented cleaning assumption; identical entries could sometimes represent legitimate repeated lines. Counts of cancellation, quantity and price exclusions overlap and must not be added together.

Positive-sales filter: Exclude invoices whose number begins with C, nonpositive quantities and nonpositive unit prices. After deduplication there are 536,641 rows; the positive-sales analysis retains **524,878**. Total exclusions from the original file are 17,031.

Customer identification: Retain otherwise valid sales with missing CustomerIDs for revenue and product units. The retained sales include **132,186 unidentified rows**. Exclude those rows only when linking purchases to individual customers.

Product descriptions: Group by StockCode, use the first available description and label an absent description "Description unavailable." Descriptions can vary, so they do not define product identity.

Revenue: Calculate Quantity x UnitPrice in GBP. Positive-sales revenue is measured before return and cancellation deductions; it is not net revenue or profit.

Evidence: [analyze_data.py](#)

Steps

2. Customer purchasing and country sales

Country rankings use the prepared positive-sales totals. Missing CustomerIDs are retained in revenue. EIRE is preserved as the source geographic label.

Country	Revenue (GBP)	Identified customers / warning	Marketing recommendation
United Kingdom	9,001,744.09	3,920 / No	Test a campaign based on recorded product purchases and measure incremental sales.
Netherlands	285,446.34	9 / No	Review recorded purchase frequency and product mix to design a small marketing test.
EIRE	283,140.52	3 / Yes	Review revenue concentration and unidentified purchases before increasing marketing spend.

Limitation: These records do not establish national demand or customer motives. The below-five warning is a descriptive teaching rule, not a statistical test. Missing IDs may represent additional customers.

Purchasing frequency among identified customers

Measured customer finding	Value
Customers with a recorded ID	4,338
Customers with one qualifying purchase invoice	1,493
Customers with at least two qualifying invoices	2,845 (65.58%)
Median distinct purchase invoices per identified customer	2

Frequency counts **distinct InvoiceNo values** per CustomerID after the positive-sales cleaning. The percentage describes repeat purchasing among identified customers within this period; it is not a retention rate for all customers.

Recommendation: Review purchase intervals and product choices among identified repeat customers before designing and measuring a marketing test.

Evidence: results/customer_purchase_summary.csv; results/country_revenue.csv; results/country_coverage.csv.

Steps

3. Product demand and inventory decisions

Here, product demand means **recorded positive units sold**, grouped by StockCode. The five products below lead under that definition. Signed units are shown as a separate cancellation-inclusive review.

Product (StockCode and description)	Positive units	Signed units	Inventory recommendation
23843 - PAPER CRAFT , LITTLE BIRDIE	80,995	0	Review transaction reversals before considering replenishment; the recorded signed balance is zero.
23166 - MEDIUM CERAMIC TOP STORAGE JAR	78,033	3,539	Review the large positive/signed-unit difference, current stock and supplier lead times before replenishment.
22197 - SMALL POPCORN HOLDER	56,898	56,427	Review the positive/signed-unit difference, current stock and supplier lead times before replenishment.
84077 - WORLD WAR 2 GLIDERS ASSTD DESIGNS	54,951	53,751	Review the positive/signed-unit difference, current stock and supplier lead times before replenishment.
85099B - JUMBO BAG RED RETROSPOT	48,371	47,256	Review the positive/signed-unit difference, current stock and supplier lead times before replenishment.

Transaction check: Product 23843 has one positive purchase row for 80,995 units. A same-day cancellation records -80,995 units with the same price and CustomerID, leaving a recorded signed balance of zero.

Limitation: Signed quantities come from deduplicated records with positive prices, including negative quantities. They do not prove that every reversal has been linked to its original purchase. Stock levels, supplier lead times and sales lost through stockouts are unavailable, so exact reorder quantities cannot be justified.

Evidence: [results/top_five_cancellation_review.csv](#)

Steps

4. Monthly trends and competitor evidence

Monthly revenue uses the same positive-sales definition. Percentage change equals (current revenue - previous revenue) / previous revenue x 100, and is reported only for consecutive months declared complete in the prepared summary.

Measured finding	Month	Supporting value
Largest qualifying month-to-month increase	May 2011	+43.27%
Largest qualifying month-to-month decrease	April 2011	-25.03%
Highest complete-month positive revenue	November 2011	GBP 1,503,866.78
Incomplete final month	December 2011	GBP 637,790.33; through Dec. 9

Planning recommendation: Review inventory timing around the recorded higher-revenue months using current stock and supplier lead times. The partial December total cannot establish a full-month decline. This period does not prove recurring seasonality or a Christmas-related cause.

Competitor research: acknowledge the missing evidence

Research question	Evidence-based conclusion / additional data needed
Does this retailer outsell competitors?	Cannot be determined. Obtain competitor sales for matching dates, product categories and countries, with consistent currencies, channels and treatment of returns and cancellations.
What is its market share?	Cannot be calculated. Obtain total-market sales covering the same scope and revenue definition.
How many retailers operate in each country?	Country sales groups do not count retailers. Obtain dated business-register or industry data with a consistent retailer definition and a distinction between companies and store locations.
Can CustomerIDs count competitors?	No. Customer identifiers do not establish competitor identities or business classifications.

Recommendation: Obtain comparable competitor evidence before making competitive-positioning claims. Missing competitor evidence must not be treated as zero sales.

Evidence: results/monthly_revenue.csv; results/country_revenue.csv; results_v3/verified_reports/competitor_brief.md.

Steps

5. Prompt engineering and broader training tasks

Prompts specify the analyst role, supplied evidence, requested output and limits on claims. Earlier country-brief validation tests produced 2/6 strict matches for minimal prompts, 3/6 for format prompts and 4/6 for grounded prompts. These are development results, not fresh test scores.

Task family	Evidence-reporting requirement	Train / val / test
Consumer behavior	Report supplied repeat-customer percentages and limit them to identified customers.	128 / 32 / 32
Country coverage	Apply the customer-count warning and preserve geographic labels and scope.	128 / 32 / 32
Market trends	Interpret supplied changes; distinguish complete from partial periods.	128 / 32 / 32
Product demand	Distinguish positive units and signed balances; require inventory information.	128 / 32 / 32
Competitor research	Compare matched fictional evidence or explain why the comparison is unsupported.	128 / 32 / 32

v3 split	Prompt examples	Underlying scenarios
Training	640	320
Validation	160	80
Test	160	80

All v3 examples are **explicitly fictional**, separate from real retailer findings. Several families include figures already computed by Python; the model is tested on evidence reporting and interpretation, not independent arithmetic.

Each scenario has two instruction variations kept in the same split. Assembly checks found no shared source groups or identical evidence across splits, and SHA-256 fingerprints were recorded. Splits still share templates and reasoning patterns; the paired variations are related observations.

Examples and prompt variations appear in Appendix A. Source: training_data_v3/manifest.json and the train, validation and test JSONL files.

Steps

6. Investigate model loading and fine-tune locally

The original submission did not record a model revision. Diagnosis used the locally cached revision; Transformers 5.18.0 loaded a tied-embedding setting that differed from the raw config. Forcing the flag alone produced missing embedding weights and unusable output.

The revised run pins **Transformers 4.57.6** and the revision below. Loading checks found no reported missing, unexpected or mismatched keys; tie_word_embeddings remained False and input/output weights had distinct storage. These checks address the observed configuration problem without proving the original run's revision or eliminating reasoning failures.

Recorded v3 setting	Value
Model	google/flan-t5-small
Revision	0fc9ddf78a1e988dac52e2dac162b0ede4fd74ab
Hardware / PyTorch	NVIDIA GeForce RTX 5090 / 2.14.1+cu130
Epochs / batch size / seed	6 / 4 / 928
Learning rate / optimizer updates	0.00005 / 960
Initial validation loss	3.468904
Selected epoch / validation loss	6 / 0.020701

Epoch	Training loss	Validation loss
1	1.685899	0.318109
2	0.368904	0.086480
3	0.161013	0.041956
4	0.090817	0.032488
5	0.058415	0.021542
6	0.043392	0.020701

Training used AdamW, masked padding labels and input/target length checks. The checkpoint was selected by validation loss before test generation. Original and tuned test responses used identical prompts and greedy generation settings.

Evidence: [results_v3/expanded_run/training_run.json](#)

Steps

7. Compare responses and inspect business errors

Strict reference agreement ignores whitespace differences only. It measures reproduction of the supplied reference wording and facts; it is not a complete semantic accuracy measure.

Test measure	Original model	Fine-tuned v3
Strict reference matches	0/160 (0%)	132/160 (82.5%)
Scenarios with both instruction variants matching	0/80	64/80 (80%)
Empty responses	12	0

Task family	Fine-tuned strict matches
Competitor research	32/32
Consumer behavior	32/32
Country coverage	15/32
Monthly trends	21/32
Product demand	32/32

Manual review of the 28 fine-tuned mismatches

- **17 customer-warning errors:** the model sometimes reversed the below-five rule or described an incorrect count relationship.
- **Nine clear monthly-trend errors:** wrong percentage values, incorrect increase/decrease directions, or both.
- **Two ambiguous monthly answers:** a decrease was phrased using a negative decrease amount, leaving the direction unclear.

The review did not justify raising the conservative complete-answer count above 132/160. The original model's zero strict matches do **not** mean zero factual accuracy; an incomplete or differently worded answer can fail this metric.

Evaluation limit: The test is synthetic, uses related templates and often supplies computed figures. It does not demonstrate reliability on real retail briefs, new wording or autonomous arithmetic. Lower validation loss did not remove consequential errors.

Evidence: [results_v3/expanded_run/strict_reference_scores.json](#)

Steps

8. Quantitative audit and implemented safeguards

A controlled regression audit uses Germany, Brazil, Unspecified and fictional Sampleland with customer counts of 3, 4, 5 and 8. Within each count group, only the geographic label changes. The descriptive warning applies when the count is below five.

Decision measure (16 cases)	Original	Fine-tuned v3	Python rule
Correct decisions	8/16	8/16	16/16
Missed required warnings (8 opportunities)	8	8	0
False warnings (8 opportunities)	0	0	0
Unclassified decisions	0	0	0
Label accuracy gap (percentage points)	0	0	0

Geographic label	Original	Fine-tuned v3	Python rule
Germany	2/4	2/4	4/4
Brazil	2/4	2/4	4/4
Unspecified	2/4	2/4	4/4
Sampleland (fictional)	2/4	2/4	4/4

Both models answered No to every case and missed every required warning. A zero label gap reflects equal measured performance, including consistent failure; it does not prove population fairness or absence of bias. These previously reviewed cases are regression checks, not an independent population study.

Application mitigation - evaluated separately

The reporting application applies **identified_customers < 5** in Python, validates counts, computes repeat purchasing and monthly changes, preserves cancellation-inclusive quantities, and uses fixed evidence templates. It produces all five briefs and their combined document.

The rule safeguard scored 16/16 on this defined audit. That result does not change the raw model's scores or establish broad fairness. Recommendations remain labeled separately from measured findings.

Evidence: [results_v3/fairness_audit/decision_scores.json](#)

Conclusion

What the resubmission establishes

The revised project broadens the actual fine-tuning dataset to five market-research task families, improves reproducibility through pinned loading checks and dataset fingerprints, and documents a quantitative audit. The v3 model reproduces the supplied reference patterns substantially better on its synthetic test set.

The model remains unreliable for unverified business reporting. Country-warning decisions and monthly percentages still fail, even with low validation loss. Synthetic reference agreement and identical-template evaluation cannot establish general real-world performance.

The working reporting application uses prepared retailer summaries, Python calculations and fixed templates for protected findings. Its verified briefs give the marketing and inventory manager measured evidence while preserving the raw model's failures transparently.

Practical takeaways

Lesson	Resulting practice
Successful training is not sufficient evidence of correctness.	Inspect generated facts and decisions in addition to reporting validation loss.
Balanced outcomes can still be consistently wrong.	Report missed warnings and false warnings, along with group-level gaps.
Data coverage defines the scope of a business claim.	Retain legitimate unidentified sales, qualify partial periods and avoid claims about competitors without their evidence.

Further work

- Evaluate new prompt formats and independently reviewed real-data cases without using them to tune the same reported test results.
- Broaden controlled audits and verify rule behavior before using generated recommendations operationally.
- Obtain stock levels, supplier lead times and matched competitor data before making reorder or competitive-positioning decisions.

The learner should review this report against the source files and explain the data-cleaning choices, train/validation/test separation, model failures and safeguard design when presenting the resubmission.

Appendix

A. Actual v3 prompt variations

These examples are copied from training_data_v3/train.jsonl. They are fictional teaching scenarios and are not findings about the Online Retail retailer.

Country coverage - same evidence, two instructions

Instruction 1: Does the fewer-than-five-customer warning apply? Answer Yes or No, state the customer count and one limitation.

Instruction 2: Apply the descriptive customer-count rule to this evidence. State whether the warning applies, explain why and give one limitation.

Evidence: Fictional teaching scenario, not Online Retail findings. Geographic label: Sampleland. Identified customers: 2. Purchase invoices: 277. Apply the warning only when identified customers are below five, regardless of invoice count. This is a descriptive rule, not a statistical test. The records describe one retailer.

Reference target: Yes, the warning applies because there are 2 identified customers, fewer than five. These records do not establish national demand or population representativeness.

Source group: country_v3_train_000; both variations are in the training split.

Consumer behavior - same evidence, two instructions

Instruction 1: Report the supplied repeat-purchase percentage and explain which customers it describes. Answer in two sentences.

Instruction 2: Summarize repeat purchasing from the supplied evidence. Include the percentage, its population and one limitation in two sentences.

Evidence: Fictional teaching scenario, not Online Retail findings. Identified customers: 939. Customers with at least two distinct qualifying purchase invoices: 511. Repeat-purchase percentage calculated by Python: 54.42%. Qualifying sales rows without CustomerIDs: 190. Qualifying purchases are non-cancelled and have positive quantities and prices. Customers without recorded IDs are not included in the customer-level calculation.

Reference target: Among identified customers, 54.42% had at least two distinct qualifying purchase invoices. This describes repeat purchasing during the recorded period and is not a retention rate for all customers.

Source group: consumer_v3_train_000; both variations are in the training split.

Prompt contract across the project

Instructions specify what the answer must report, supply evidence explicitly, distinguish recommendations from findings and identify limits on inference. Competitor training compares labeled fictional scenarios; the real retailer brief acknowledges unavailable competitor figures.

Appendix

B. Evidence and reproducibility

Project: <https://github.com/rwnycvet/sba928-market-research/tree/sba928-resubmission>

Analysis checkpoint 3170416. The tables in this PDF are prepared from saved artifacts; they are not newly executed model tests or recreated terminal screenshots.

Saved evidence	What it supports
analyze_data.py; results/*.csv	Retail cleaning, sales, units, customer frequency and recorded monthly coverage.
training_data_v3/manifest.json; train.jsonl; validation.jsonl; test.jsonl	Split counts, scenario groups, five task families and SHA-256 fingerprints.
results_v3/expanded_run/training_run.json	Model revision, software/hardware settings, loss history and selected epoch.
results_v3/expanded_run/model_comparison.json; strict_reference_scores.json; strict_reference_review.json	Original/tuned generations, strict scores and response-level review.
results_v3/fairness_audit/responses.json; decision_scores.json	Controlled-label audit answers, error counts and safeguard comparison.
results_v3/verified_reports/retail_research_briefs.md; results_v3/evaluation.md	Five application briefs and the stated evaluation limitations.
submission_v1/; results_v2/; resubmission_plan.md	Preserved original submission and earlier diagnosis/development evidence.

Recorded dataset fingerprints (SHA-256)

train:
39fd2967e2768c194b1cb30735a4adf217338a48a0d45ccc9ef3898fb715c20b

validation:
439b333c1e34eae10a9ab88a1333ca9499d28ec8d4852ef6551ed698166373ea

test:
af210a4c55646e0850a618e3a6b1a5a5087a6a84c6bb43b730fc6f9902beff05

Run the application from the project folder

```
uv sync --locked
uv run python build_verified_briefs.py
```

Training and scoring: train_multitask_v3.py and score_v3_responses.py. Audit: run_fairness_audit_v3.py and score_fairness_audit_v3.py. Training uses the configured CUDA environment; local model weights and the Excel file are excluded from Git.